

# Unit 3 Notes

## Triangles - Properties and Congruence

**How to use this file:** Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

## Unit 3 Notes: Triangles - Properties and Congruence

### Big goal

A triangle is rigid: once enough side and angle information is fixed, its shape is forced. This unit explains relationships inside one triangle and the evidence needed to prove two triangles congruent.

### 1. Classifying triangles

Triangles can be classified by sides and by angles.

- **Scalene:** no congruent sides.
- **Isosceles:** at least two congruent sides.
- **Equilateral:** three congruent sides; each angle is  $60^\circ$ .
- **Acute:** all angles less than  $90^\circ$ .
- **Right:** one angle exactly  $90^\circ$ .
- **Obtuse:** one angle greater than  $90^\circ$ .

### 2. Triangle angle theorems

#### Triangle Sum Theorem

The interior angles of every triangle add to  $180^\circ$ :

$$m\angle A + m\angle B + m\angle C = 180^\circ.$$

#### Exterior Angle Theorem

An exterior angle of a triangle equals the sum of the two remote interior angles. If exterior angle  $\angle 1$  is outside the triangle and the remote interiors are  $\angle A$  and  $\angle B$ , then

$$m\angle 1 = m\angle A + m\angle B.$$

The **Third Angles Theorem** says that if two angles of one triangle are congruent to two angles of another triangle, then the third angles are congruent.

### 3. Isosceles triangles

In an isosceles triangle, the congruent sides are the legs, and the third side is the base. The angles touching the base are the base angles.

#### Isosceles Triangle Theorems

If two sides of a triangle are congruent, then the angles opposite those sides are congruent. Conversely, if two angles of a triangle are congruent, then the sides opposite those angles are congruent.

## 4. Triangle inequality

Three lengths form a triangle only if the sum of any two side lengths is greater than the third. For sides  $a, b, c$ :

$$a + b > c, \quad a + c > b, \quad b + c > a.$$

A shortcut for an unknown third side  $x$  with sides  $a$  and  $b$  is

$$|a - b| < x < a + b.$$

## 5. Coordinate triangle classification

Use distance to classify by sides. Use slope and the Pythagorean Theorem to classify by angles.

- If two side lengths match, the triangle is isosceles.
- If all three match, the triangle is equilateral.
- If two side slopes are negative reciprocals, the triangle is right.
- If side lengths satisfy  $a^2 + b^2 = c^2$ , the triangle is right.

## 6. Congruent triangles

Two triangles are congruent when corresponding sides and corresponding angles are congruent. Corresponding parts must be listed in the same order. If  $\triangle ABC \cong \triangle DEF$ , then  $A \leftrightarrow D$ ,  $B \leftrightarrow E$ , and  $C \leftrightarrow F$ .

| Criterion | What proves congruence                                                     |
|-----------|----------------------------------------------------------------------------|
| SSS       | Three pairs of corresponding sides are congruent.                          |
| SAS       | Two pairs of corresponding sides and the included angle are congruent.     |
| ASA       | Two pairs of corresponding angles and the included side are congruent.     |
| AAS       | Two pairs of corresponding angles and a non-included side are congruent.   |
| HL        | Right triangles with congruent hypotenuses and one pair of congruent legs. |

### Invalid shortcuts

AAA proves similarity, not congruence. SSA is not a valid congruence shortcut in general because it can create two different triangles.

## 7. Proof strategy and CPCTC

Most triangle proofs follow this rhythm:

1. Mark or state all given congruent parts.
2. Look for shared parts, called the Reflexive Property. For example,  $AC \cong AC$ .
3. Use vertical angles, parallel line angles, or isosceles theorems to find more congruent parts.
4. Choose SSS, SAS, ASA, AAS, or HL.
5. After triangles are proved congruent, use CPCTC: Corresponding Parts of Congruent Triangles are Congruent.

## 8. Midsegments and medians

A **midsegment** joins the midpoints of two sides of a triangle. It is parallel to the third side and half its length. If  $M$  and  $N$  are midpoints of two sides and  $BC = 24$ , then  $MN = 12$  and  $MN \parallel BC$ .

A **median** joins a vertex to the midpoint of the opposite side. The three medians meet at the **centroid**. The centroid divides each median in a  $2 : 1$  ratio, with the longer part from the vertex to the centroid.

## 9. Rigid motions and triangle congruence

Rigid motions explain why SSS, SAS, and ASA force congruence. If one triangle can be translated, rotated, and/or reflected to match another exactly, then every corresponding side and angle matches.

## 10. Worked examples

### Example 1: Exterior angle

If a triangle has remote interior angles  $42^\circ$  and  $73^\circ$ , the exterior angle is

$$42^\circ + 73^\circ = 115^\circ.$$

### Example 2: Isosceles algebra

Base angles are congruent, so if they measure  $4x - 3$  and  $2x + 25$ , then

$$4x - 3 = 2x + 25 \Rightarrow 2x = 28 \Rightarrow x = 14.$$

Each base angle is  $53^\circ$ , and the vertex angle is  $180 - 106 = 74^\circ$ .

### Example 3: Congruence proof skeleton

Given  $AB \cong DE$ ,  $AC \cong DF$ , and  $\angle A \cong \angle D$ , the angle is included between the two sides. Therefore  $\triangle ABC \cong \triangle DEF$  by SAS.

### Example 4: Midsegment

If a midsegment is  $3x + 1$  and the third side is 32, then the midsegment is 16. Solve  $3x + 1 = 16$ , so  $x = 5$ .

## Mastery checklist

You are ready for Unit 3 when you can solve triangle angle equations, classify triangles from coordinates, explain isosceles relationships, choose valid congruence criteria, write short proofs using CPCTC, and use midsegment and centroid facts.